

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

CE-5003-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Structural Analysis - II

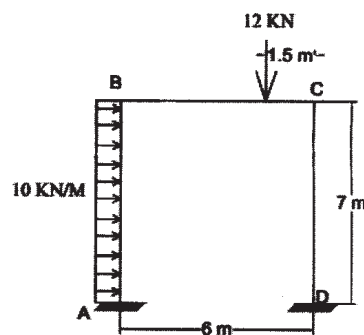
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Any missing data may be suitably assumed, if any.

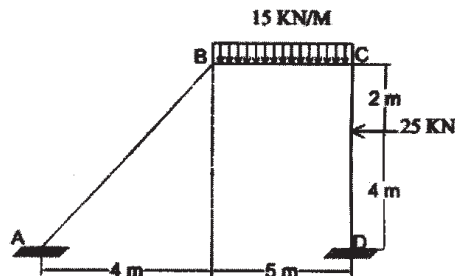
1. Analyse the frames using Kani's method and draw the BMD.

14



2. Analyse the frame using moment distribution method and draw the BMD.

14

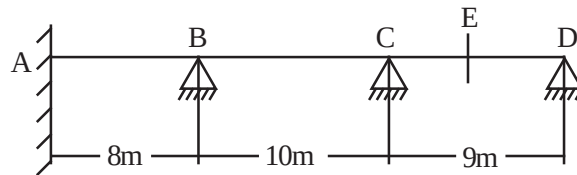


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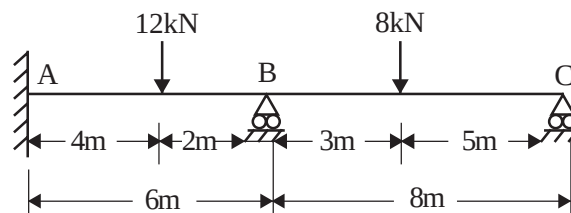
PTO

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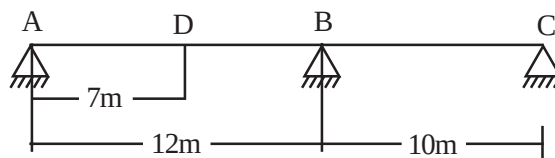
3. Determine the equation for the influence line for shear at E of the following statically indeterminate beam. The load moves from A to D. Draw the influences line diagram for the shear at E. At a suitable intervals plot diagram. 14



4. Analyze the continuous beam by using flexibility method of matrix if the supports B sinks by 10mm and $EI = 15 \times 10^3$. Draw the BMD. 14



5. Define the following
- i) Shape factor 3
 - ii) Working load 2
 - iii) Load factor 2
 - iv) Factor of safety 2
 - v) Plastic moment capacity 3
 - vi) Collapse load 2
6. Using Muller Breslau principle, draw the influence line for bending moment at the point D of the span AB of the continuous beam ABC. Determine the influence line ordinates at suitable intervals and plot them 14

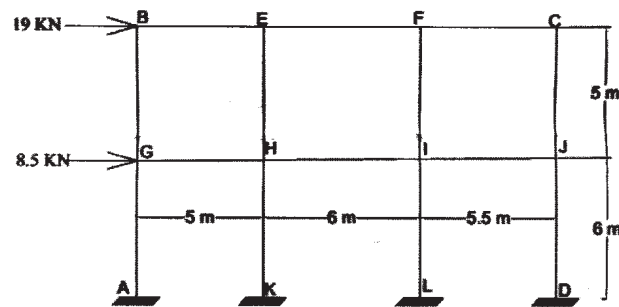


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7. Analyse the building frame using cantilever method assume if the data is missing. Sketch the BMD. 14



8. Analyse the continuous beam by using stiffness method of matrix. Draw the BMD. 14

